SS3 to opal: eastern Pacific bigeye tuna

Bridging the IATTC SAC-14 exploratory model, a quarterly Stock Synthesis model with growth cessation, to opal

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1 Overview

This document bridges an exploratory Stock Synthesis [1] assessment model of bigeye tuna (*Thunnus obesus*) in the eastern Pacific Ocean (EPO), prepared by IATTC staff for the 14th meeting of the IATTC Scientific Advisory Committee [2], to opal. Any model runs here are preliminary explorations and should not be used *as is* for management advice.

The baseline SS3 model is itself a simplification. The models in the IATTC's risk analysis for EPO bigeye have two sexes and time-varying selectivity; the model used here has one sex and time-invariant selectivity, so that its structure is close to what opal can represent. Its results are not those of the risk-analysis models, and differences between SS3 and opal found here say nothing about how opal would compare with them.

- **The model is quarterly.** Each SS3 "year" is a quarter (17-204, taken as 1975 Q1 to 2021 Q4) and ages are quarters (0-40). opal takes the same layout: 188 time steps of one season and 41 ages.
- **One SS3 bridge step.** The supplied model (B0) estimates a regime offset on the initial equilibrium recruitment, which opal does not have; B1 removes it (Section 3). Growth needs no step: SS3′s growth cessation model [3] and its SD-linear growth variability are both in opal, and the conversion reproduces SS3′s length-at-age and SD-at-age exactly below the plus group (Section 4.2).
- **The inputs reconcile; the dynamics differ.** The converted inputs reconcile with SS3 within the tolerances of Table 3, and opal at SS3′s parameter values matches an independent replay of the converted inputs, so the gap between opal and SS3 lies in the dynamics. opal removes each quarter's catch at the start of the quarter, as a harvest rate, while SS3 removes it through the quarter. With M of 0.48 per quarter at age 0 and

large purse-seine catches of young fish, that leaves far more fish alive in opal (Section 4.4). Catch timing is the largest difference between the two models for this stock.

- **opal fits** (Section 5) estimate the scale and recruitment with SS3's selectivity held fixed (O1) and with opal's own double normals (O2).
- **Initial F** (Section 6). SS3 fixes the initial F of the two fleets with an initial equilibrium catch (LL-n-A2 and OBJ-A5). Estimating them adds nothing with the regime offset on (B0-F). Without it (B1-F) the initial F of OBJ-A5 takes over much of the offset's job and brings SS3 back close to B0, but in opal (O1-F) it stays near 0 and changes almost nothing.

2 The model

Table 1: Fisheries and data in opal (SS3 fleet numbers are kept). Units: catch in weight (1) or numbers (2). Compositions: the variance adjustment is opal's divisor of the input sample size, the inverse of SS3's multiplier.

opal fishery	SS3 fleet	name	catch units	compositions	variance adjustment
1	1	LL-n-A1	2	none	1.00
2	2	LL-n-A2	2	size (10 cm)	1.25
3	3	LL-n-A3	2	size (10 cm)	1.69
4	4	LL-n-A4	2	size (10 cm)	2.50
5	5	LL-n-A5	2	size (10 cm)	2.50
6	6	LL-n-A6	2	size (10 cm)	3.23
7	7	LL-n-A7	2	size (10 cm)	2.40
8	8	LL-w-A1	1	none	1.00
9	9	LL-w-A2	1	none	1.00
10	10	LL-w-A3	1	none	1.00
11	11	LL-w-A4	1	none	1.00
12	12	LL-w-A5	1	none	1.00
13	13	LL-w-A6	1	none	1.00
14	14	LL-w-A7	1	none	1.00
15	15	OBJ-A1	1	length (2 cm)	4.13
16	16	OBJ-A2	1	length (2 cm)	8.33
17	17	OBJ-A3	1	length (2 cm)	35.71
18	18	OBJ-A4	1	length (2 cm)	8.33
19	19	OBJ-A5	1	length (2 cm)	37.88
20	20	OBJ-Disc	1	none	1.00
21	21	NOA-A1	1	none	1.00
22	22	NOA-A2	1	none	1.00
23	23	LL-Survey	2	size (10 cm)	1.25

- **Fisheries.** Longline fisheries in seven areas, with catch in numbers (LL-n) and in weight (LL-w; their selectivity mirrors the LL-n fishery of the same area); floating-object purse seine in five areas (OBJ) and its discards (OBJ-Disc, age-based selectivity of 1 at ages 1-3 quarters); unassociated purse seine in two areas (NOA, mirroring OBJ-A1 and OBJ-A5).

- **Index.** One longline index (LL-Survey, numbers, lognormal), with its own double-normal selectivity and SS3's floating catchability.
- **Compositions.** OBJ length compositions on 2 cm bins (20-198 cm); the longline fisheries' and the index's compositions are SS3 generalised size compositions on 10 cm bins (60-190 cm). In opal they are two composition bin schemes predicted from the same 2 cm population grid. SS3 gives zero weight to the NOA length compositions, LL-n-A1's size compositions and a handful of conditional age-at-length records, which are left out.
- **Biology.** Age-specific M (0.48 per quarter at age 0, falling to 0.10 at age 13 and rising to 0.14 from age 26), growth cessation, length-based maturity, spawning output proportional to weight (Nsexes = −1, female fraction 0.5), Beverton-Holt recruitment with steepness 1 and sigma_R 0.6.

3 The SS3 bridge

Table 2: The SS3 bridge. Negative log-likelihoods; SSB in tonnes; depletion and F/F_{MSY} in the last quarter (2021 Q4).

step	n par	NLL	survey	length	size freq.	recruitment	ln R0	regime offset	SSB0	depletion	F/Fmsy	max gradient
B0 SAC-14 model	264	271.20	−261.19	220.85	327.10	−15.58	9.449	−0.409	371,192	0.221	−1.57	1.8e-03
B1 no regime offset	263	274.60	−261.35	220.60	329.12	−13.79	9.457	0.000	374,298	0.249	−1.65	2.0e-02

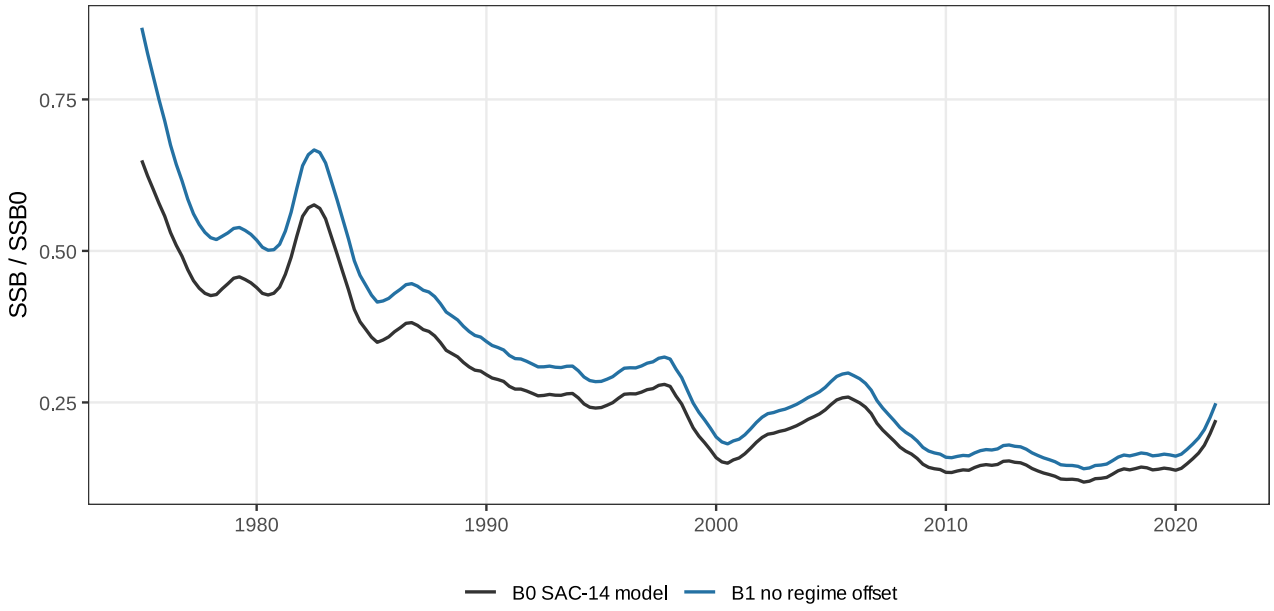


Figure 1: Spawning biomass relative to unfished in the two SS3 bridge models.

The supplied files were run with an SS3 3.30.22 beta build. Re-run with SS3 3.30.22.1 from the supplied parameters, B0 reproduces the supplied likelihood exactly (271.196) and re-estimation stays there.

B0 starts the population from an equilibrium whose recruitment is 34% below the stock-recruitment curve (a regime offset of −0.41 on ln R_0 for block year 16, the quarter before the model starts). The model begins in 1975 although longline fishing on the stock goes back decades (the data file carries catches for 1971-1974 that SS3 ignores, and the initial F is fixed at almost 0), and the offset is what lets the 1975 population start below its unfished level: it scales every initial age class by exp(-0.41). SS3's lambda on the regime penalty is 0, so the offset is informed by the data alone; SS3 estimates it at −0.41 (SE 0.16).

B1 removes it. The initial equilibrium recruitment is then tied to the unfished stock-recruitment relationship, while the 24 penalised early recruitment deviations can still alter the initial age structure; they take over part of the offset's job (their mean is −0.24 in B1 against 0.00 in B0). The likelihood worsens by 3.4 units, and the difference does not wash out: B1′s spawning biomass is 35% above B0′s in 1975 Q1, 16% to 23% above it from 1980 to 2010, and 14% above it in 2021 Q4 (depletion 0.249 against 0.221). opal is converted from B1, so the comparison in the rest of this document is with a model whose trajectory differs from the supplied one by this much (Section 7).

4 Converting B1 to opal

`config/epo-bet-b1.yaml` records the choices the SS3 files do not determine. The opal model has:

- 188 quarterly time steps, one season each, ages 0-40 quarters;
- 23 fisheries: SS3′s 22 fishing fleets in SS3 order and one fishery for the index;
- SS3′s age-basis biology (M, maturity, and spawning output) from the endgrowth table at the start of the quarter, and SS3′s *mid*-quarter weight-at-age, which opal uses to turn catch in weight into numbers (Section 4.4);
- SS3′s parameter values: B_0 (opal's scale parameter, SS3′s SSB_0), h, $sigma_R$, the main and late recruitment deviations, the 24 early deviations and the first quarter's recruitment as the initial age structure, the initial F of LL-n-A2 and OBJ-A5, and SS3′s growth.

4.1 Reconciliation

Table 3: Reconciliation with SS3 (exact comparisons). The 22 catch totals are summarised in one row.

Quantity	n	max relative difference	pass
catch totals, 22 fishing fleets	22	0.0e+00	TRUE
CPUE observation count (kept SS3 index fleets)	1	0.0e+00	TRUE
LF total input sample size	1	0.0e+00	TRUE
LF observation rows	1	0.0e+00	TRUE
unfished spawning potential per recruit	1	1.8e-06	TRUE
total weight summed over ages (Wt_Mid)	1	0.0e+00	TRUE
mean length-at-age, SS3 growth cessation (max relative error below the plus group)	1	4.5e-06	TRUE
SD of length-at-age, SS3 CV_Growth_Pattern 2 (max relative error below the plus group)	1	9.6e-07	TRUE

All 38 structural checks pass. The unfished spawning potential per recruit reconciles to 1.8e-06, which confirms the units of M, the female fraction and the plus group in one number.

4.2 Growth

SS3′s growth cessation model (GrowthModel 8) is Eq. 2 of M. N. Maunder *et al.* [3], which opal implements as `growth_model = "cessation"`. The parameters map directly:

SS3	Value	opal
L at age 0 (AFIX = 0)	22.79 cm	`L0`
rmax (labelled VonBert_K)	10.2127 cm per quarter	`rmax`
Cessation	0.1811	`k_gc`

SS3	Value	opal
Linf (Growth_Age_for_L2 = 999)	191.628 cm	`t50` = 16.248 quarters, as SS3 computes it from Linf
SS3 age 0 = opal position 1		`age_offset` = −1

SS3′s CV_Growth_Pattern 2 makes the SD of length-at-age linear in mean length, from 3.53 cm at L_0 to 9.48 cm at Linf; opal's `growth_sd_model = "sd"` does the same. Linf is not a modelled length, so opal's upper reference is the mean length at the oldest age, with the SD SS3′s line takes there (9.453 cm).

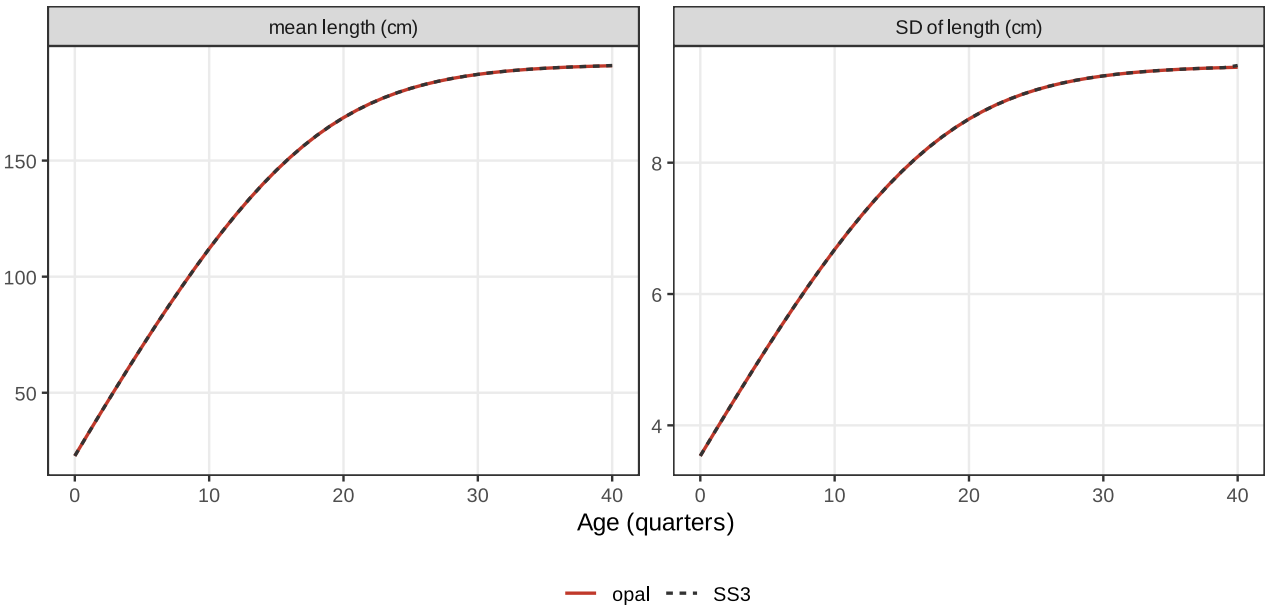


Figure 2: Mean and SD of length-at-age: SS3 (Report.sso, start of the quarter) and opal.

Below the plus group opal's length-at-age matches SS3′s to 4.5e-06 and its SD to 9.6e-07 (relative), the precision of Report.sso. At the plus group SS3 averages length over the ages it accumulates (190.96 cm against opal's 190.87 cm) and holds the SD at SD_old itself.

4.3 Selectivity

SS3′s selectivity is size-based, like opal's, so its double normals carry over rather than being fitted. Two details are converted:

- SS3 places the end of the plateau relative to the midpoint of its last *population* length bin (221 cm), opal relative to its last data length bin (199 cm). opal's top parameter is re-solved so the plateau ends where SS3′s does.
- Where SS3 sets a start or end parameter to −999 (no rescaling of that limb), opal's end value is set to the value SS3′s unscaled limb takes at the first or last bin.

OBJ-Disc's selectivity is on age in SS3 (1 at ages 1-3 quarters, 0 elsewhere); in opal it is an age-based double normal fitted to that and held fixed.

Table 4: Largest difference between opal's starting selectivity and SS3's: at length for the length-based fisheries (mirrors take their source fleet's parameters), at age for OBJ-Disc.

opal fishery	name	parameters from	max |diff|
1	LL-n-A1	1	1.3e-05
2	LL-n-A2	2	9.4e-06
3	LL-n-A3	3	1.6e-06
4	LL-n-A4	4	5.8e-05
5	LL-n-A5	5	5.2e-06
6	LL-n-A6	6	3.0e-06
7	LL-n-A7	7	4.5e-06
8	LL-w-A1	1	1.3e-05
9	LL-w-A2	2	9.4e-06
10	LL-w-A3	3	1.6e-06
11	LL-w-A4	4	5.8e-05
12	LL-w-A5	5	5.2e-06
13	LL-w-A6	6	3.0e-06
14	LL-w-A7	7	4.5e-06
15	OBJ-A1	15	2.7e-06
16	OBJ-A2	16	1.4e-05
17	OBJ-A3	17	2.6e-06
18	OBJ-A4	18	4.6e-04
19	OBJ-A5	19	3.0e-06
20	OBJ-Disc	age-based fit	1.4e-03
21	NOA-A1	15	2.7e-06
22	NOA-A2	19	3.0e-06
23	LL-Survey	23	1.0e-05

Every fleet's SS3 age selectivity (pattern 10) is 0 at age 0. opal applies its length curves to every age through the PLA, so the OBJ fisheries select some age-0 fish in opal that SS3 does not; at the start of the quarter an age-0 fish is 22.8 cm. SS3 also evaluates fishery selectivity on the mid-quarter size distribution, opal at the start of the quarter. Both show in opal's starting selectivity-at-age:

Table 5: Largest difference between opal's starting selectivity-at-age (O2 before fitting) and SS3′s, and opal's selectivity at age 0, by fishery.

fishery	name	max |diff|	opal at age 0
1	LL-n-A1	0.117	0.000
2	LL-n-A2	0.073	0.000
3	LL-n-A3	0.133	0.000
4	LL-n-A4	0.120	0.000
5	LL-n-A5	0.088	0.000
6	LL-n-A6	0.080	0.000
7	LL-n-A7	0.121	0.000
8	LL-w-A1	0.117	0.000
9	LL-w-A2	0.073	0.000
10	LL-w-A3	0.133	0.000
11	LL-w-A4	0.120	0.000
12	LL-w-A5	0.088	0.000
13	LL-w-A6	0.080	0.000
14	LL-w-A7	0.121	0.000
15	OBJ-A1	0.322	0.020
16	OBJ-A2	0.280	0.013
17	OBJ-A3	0.237	0.026
18	OBJ-A4	0.264	0.004
19	OBJ-A5	0.140	0.010
20	OBJ-Disc	0.001	0.001
21	NOA-A1	0.322	0.020
22	NOA-A2	0.140	0.010
23	LL-Survey	0.084	0.000

4.4 opal at SS3′s values

O0 runs opal at SS3′s parameter values with SS3′s own selectivity-at-age. Nothing is estimated, so any difference from SS3 comes from the two models' dynamics. To find out which part of the dynamics, this section compares both models with a set of *replays*.

4.4.1 How the replays work

A replay is a short population projection written in plain R (in `scripts/02-convert-fit-epo-bet-opal.r`), outside both opal and SS3. It estimates nothing. It takes a fixed set of inputs, all at SS3 B1′s values, and steps the population forward one quarter at a time:

- the numbers-at-age in 1975 Q1 (opal's, which equal SS3′s to 0.00% in spawning biomass);
- the recruitment in each quarter, $R_0 \exp(\text{dev}_y - b_y \sigma_R^2 / 2)$, with SS3′s deviations dev_y and bias adjustment b_y (steepness is 1, so recruitment does not depend on spawning biomass);
- M-at-age, spawning output-at-age, and SS3′s selectivity-at-age $s_{f,a}$ for each fleet f;
- the observed catch $C_{f,y}$ of each fleet in each quarter.

In each quarter the replay records spawning biomass, Σ_a N_a × spawning output$_a$, then takes the quarter's catch out of the numbers-at-age N_a, applies natural mortality, moves every age class up one quarter (the oldest stays in the plus group), and adds the next quarter's recruits. The replays differ only in how the catch comes out, which is the one step where opal and SS3 differ:

- **Start of quarter** (opal's `do_dynamics()`): each fleet's catch is a harvest rate on the numbers at the start of the quarter, taken before any natural mortality, and the survivors then die at rate M:

$$h_{f,a} = \frac{C_f \, s_{f,a}}{\sum_{a'} N_{a'} \, s_{f,a'} \, w_{f,a'}}, \qquad N'_a = N_a \left(1 - \sum_f h_{f,a} \right) e^{-M_a}.$$

- **Mid-quarter** (Pope's approximation): half the quarter's natural mortality, then the same harvest rate taken from what is left, then the other half: $N'_a = \left(N_a e^{-M_a/2} \right) \left(1 - \sum_f h_{f,a} \right) e^{-M_a/2}$, with $h_{f,a}$ computed from $N_a e^{-M_a/2}$.
- **Through the quarter** (the Baranov catch equation): fishing and natural mortality act together all quarter, and each fleet's F_f is solved so that its predicted catch matches the observed one, which is what SS3′s hybrid F method does:

$$C_f = \sum_a \frac{F_f \, s_{f,a}}{Z_a} N_a \left(1 - e^{-Z_a} \right) w_{f,a}, \qquad Z_a = M_a + \sum_f F_f \, s_{f,a}, \qquad N'_a = N_a e^{-Z_a}.$$

For fleets whose catch is in weight, $w_{f,a}$ turns tonnes into numbers. It is one of three weights: start-of-quarter weight; mid-quarter weight (the one vector opal uses here); or each fleet's own catch weight-at-age from SS3 (Report.sso's `bodywt`). SS3′s weight follows growth through the quarter and the size of fish each fleet selects within an age. For fleets whose catch is in numbers, w = 1.

The replays serve two purposes:

- **A check on the conversion.** The start-of-quarter replay with opal's weight does exactly what opal does. If opal matches it, opal is using the converted inputs as intended. Figure 3 shows this ("replay, opal timing", from `ss3_bridge_replica()` in `shared/ss3-to-opal/`).
- **Attribution.** Every replay uses the same inputs and differs only in the catch step, so the differences between them (Table 6, Figure 4) are caused by when the catch is removed and at what weight. Nothing else differs, including growth, M, recruitment and selectivity. The replay that uses SS3′s catch equation and SS3′s fleet catch weights reproduces SS3 almost exactly. So with the same inputs, the catch step is the only thing separating opal's dynamics from SS3′s.

The replays use SS3′s selectivity-at-age, so they test the dynamics, not opal's own selectivity curves (that is O2, Section 5).

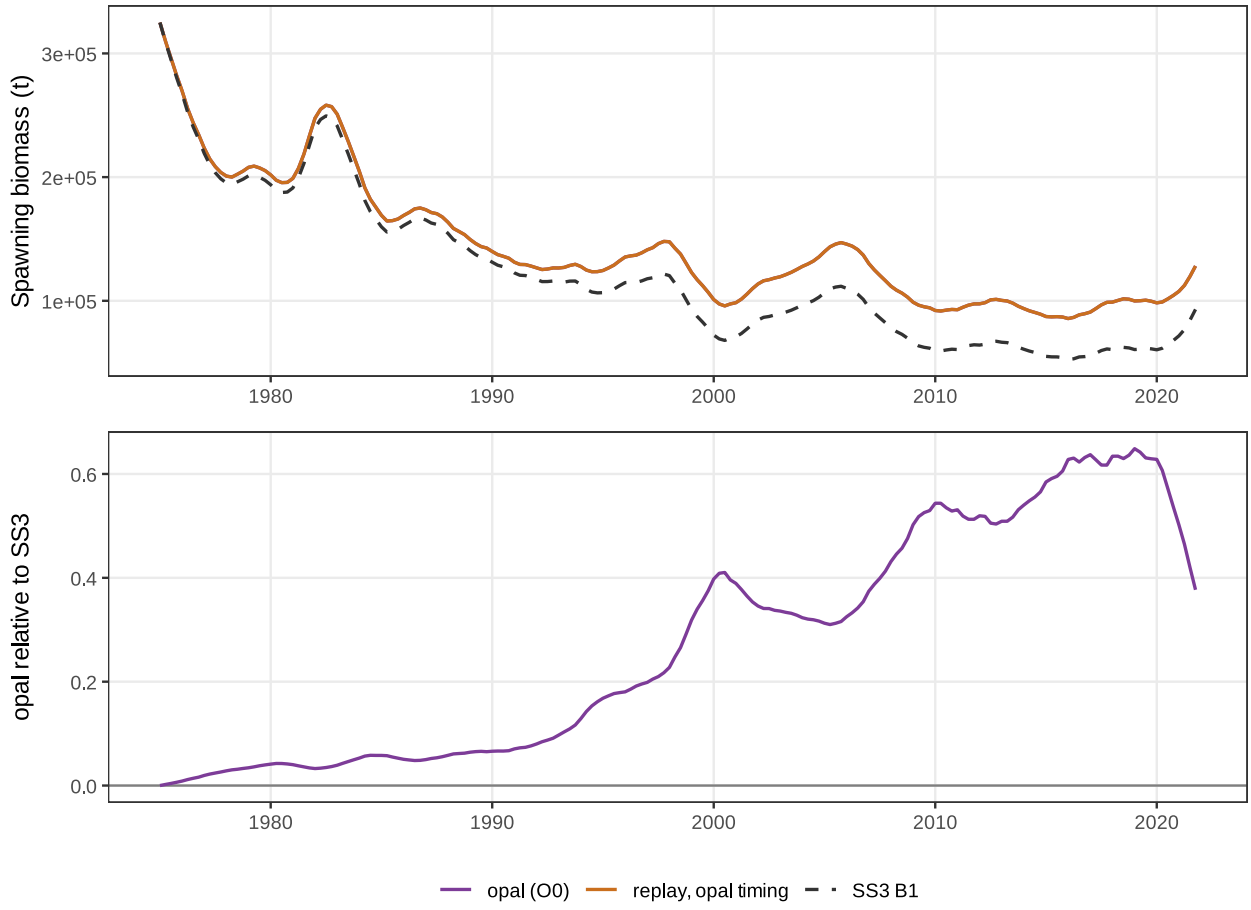


Figure 3: O0: opal at SS3′s values against SS3 B1 (top: spawning biomass; bottom: relative to SS3). 'Replay, opal timing' is the start-of-quarter replay of Section 4.4.1 with opal's weight, which opal matches exactly.

The first quarter agrees to 0.00%: the initial age structure is converted directly. opal then drifts above SS3, to 38% by 2021 Q4. opal's spawning biomass matches the replay of the converted inputs to 3e-07 (relative), which checks that opal uses the converted inputs as intended; the difference from SS3 is in the dynamics. Two things account for it: when the catch is taken within the quarter, and the weight at which catch in weight is turned into numbers.

Table 6: The replays of Section 4.4.1: spawning biomass relative to SS3 at SS3′s parameter values, by when the catch is removed and the weight used to turn catch in weight into numbers. Every replay starts from opal's initial numbers-at-age (O0) and uses SS3′s selectivity-at-age and recruitment, so they differ only in the catch step. The second row is opal's own calculation (it matches O0).

removal	weight	2001-2021 mean	2021 Q4	largest |diff|
start of quarter	start-of-quarter	−46.1%	−62.2%	84.80%
start of quarter	mid-quarter (opal here)	48.4%	37.7%	64.87%
start of quarter	SS3 fleet catch weight	43.1%	33.9%	58.04%
mid-quarter (Pope)	SS3 fleet catch weight	−3.6%	−4.8%	7.55%
through the quarter (Baranov, SS3)	SS3 fleet catch weight	0.0%	0.0%	0.00%

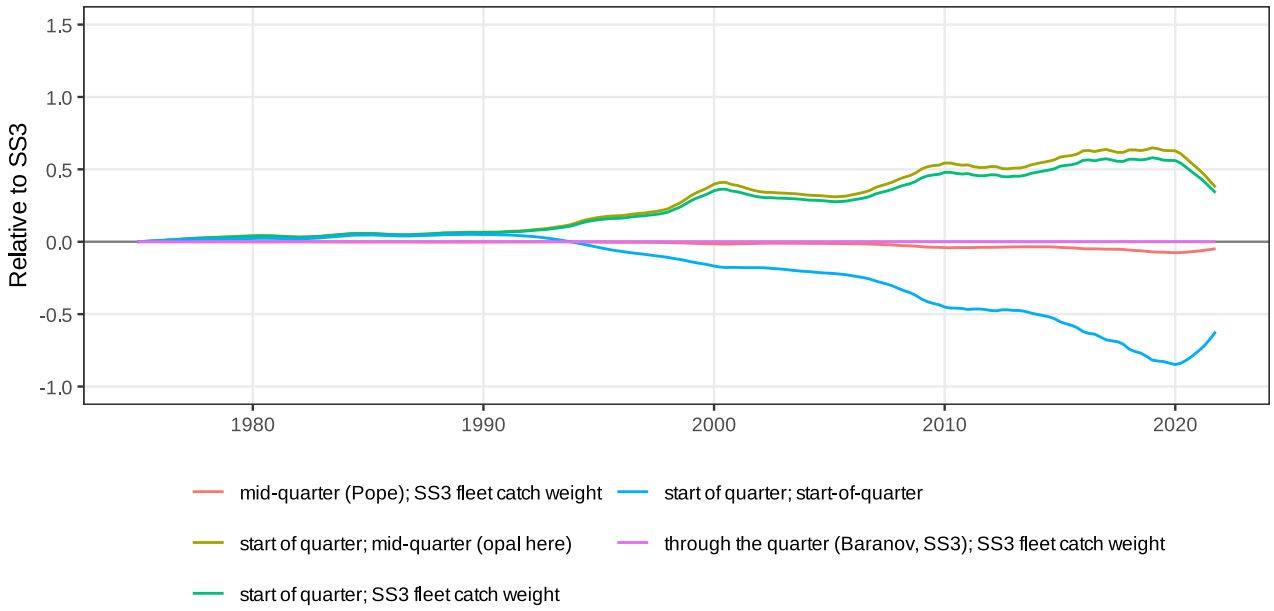


Figure 4: The replays of Table 6 relative to SS3.

- **Catch weight.** SS3 turns catch in weight into numbers with each fleet's catch weight-at-age, which follows the fish's growth through the quarter and the fleet's selection within an age. Bigeye grow fast: an age-0 fish weighs 71% more at mid-quarter than at the start. opal has one weight-at-age vector. At the start-of-quarter weight it removes so many young fish for the same tonnage that the population collapses at SS3′s parameters; the mid-quarter weight (used here) and SS3′s fleet catch weights give similar results.
- **Catch timing.** opal removes each quarter's catch at the start of the quarter, before natural mortality. SS3 removes it through the quarter (hybrid F), so part of the catch is fish that would otherwise have died. For the same catch opal leaves more fish alive, and with M of 0.48 per quarter at age 0 and the OBJ fisheries catching mostly fish under a year old, the difference accumulates. Removing the catch at mid-quarter (Pope's approximation) brings the replay to within 8% of SS3 over the series; SS3′s catch equation with SS3′s fleet catch weights reproduces SS3 to 1e-05 (relative; the largest difference in any quarter).

5 opal fits

Growth, h, sigma_R, M, the biology and the initial F stay at SS3′s values (except in O1-F, Section 6), as they are fixed in SS3 (except SD_young, held at SS3′s estimate); opal's default priors are switched off, as SS3 has no parameter priors. Both fits estimate B_0, the survey catchability, the main and late recruitment deviations and the initial age structure; the last deviation, which belongs to the quarter after the model, is fixed at 0.

- **O1** holds SS3′s selectivity-at-age.
- **O2** estimates opal's own length-based double normals, started at SS3′s curves, for the parameters SS3 estimates (the peak, top and widths of LL-n-A2 to A7, OBJ-A1 to A5 and the index); mirrors share their source's parameters. It starts from O1′s estimates.
- **O1-noComp** is O1 with the compositions switched off (lambda 0), so the index and the recruitment and initial-deviation penalties inform it.

Table 7: Fits. The objective values are opal's and are not comparable with SS3′s likelihood (different constants, and opal's compositions are predicted differently). Depletion is SSB / SSB0 in 2021 Q4. CPUE, compositions and recruitment are the negative log-likelihood terms opal reports (O1-noComp's composition term is reported but not in its objective, lambda 0); the objective also contains the initial-deviation penalty, and 'unreported' is the objective minus the sum of every weighted term opal reports (see Section 7).

model	npar	objective	max gradient	Hessian PD	B0 (t)	SSB 2021 Q4 (t)	depletion	CPUE	compositions	recruitment	unreported
SS3 B1	263				374,298	93,051	0.249				
O0	0	6,520.79			374,298	128,126	0.342	32.4	4,043.3	107.9	2,318.1
O1	214	6,369.85	1.9e-11	TRUE	427,626	142,580	0.333	−88.9	4,032.6	106.7	2,302.4
O2	262	6,169.84	5.4e-02	FALSE	451,116	159,711	0.354	−95.5	3,753.2	103.8	2,391.3
O1-noComp	214	−6.54	3.6e-09	TRUE	382,806	113,803	0.297	−127.9	4,210.0 (not in objective)	103.8	−0.0
O1-F	216	6,369.83	3.5e-06	TRUE	427,943	142,363	0.333	−88.9	4,032.6	106.7	2,302.4

O1 converges and passes opal's fit check. With SS3′s selectivity fixed, opal's own optimum puts B_0 at 427,626 t (14% above SS3′s) and the 2021 Q4 depletion at 0.333, against SS3′s 0.249. That is close to where opal already sits at SS3′s values (O0, 0.342): re-estimating the scale and recruitment does not undo the catch-timing difference of Section 4.4, because in opal's dynamics the same catches deplete the stock less. Without the compositions (O1-noComp), the index and the recruitment and initial-deviation penalties give a depletion of 0.297 (B_0 382,806 t), so the compositions push O1 towards a larger, less depleted stock.

O2 does not converge (maximum gradient 5.4e-02) and is shown for completeness only. Freeing opal's selectivity improves the fit to the compositions by 279 units over O1, and its depletion of 0.354 should not be read as an estimate.

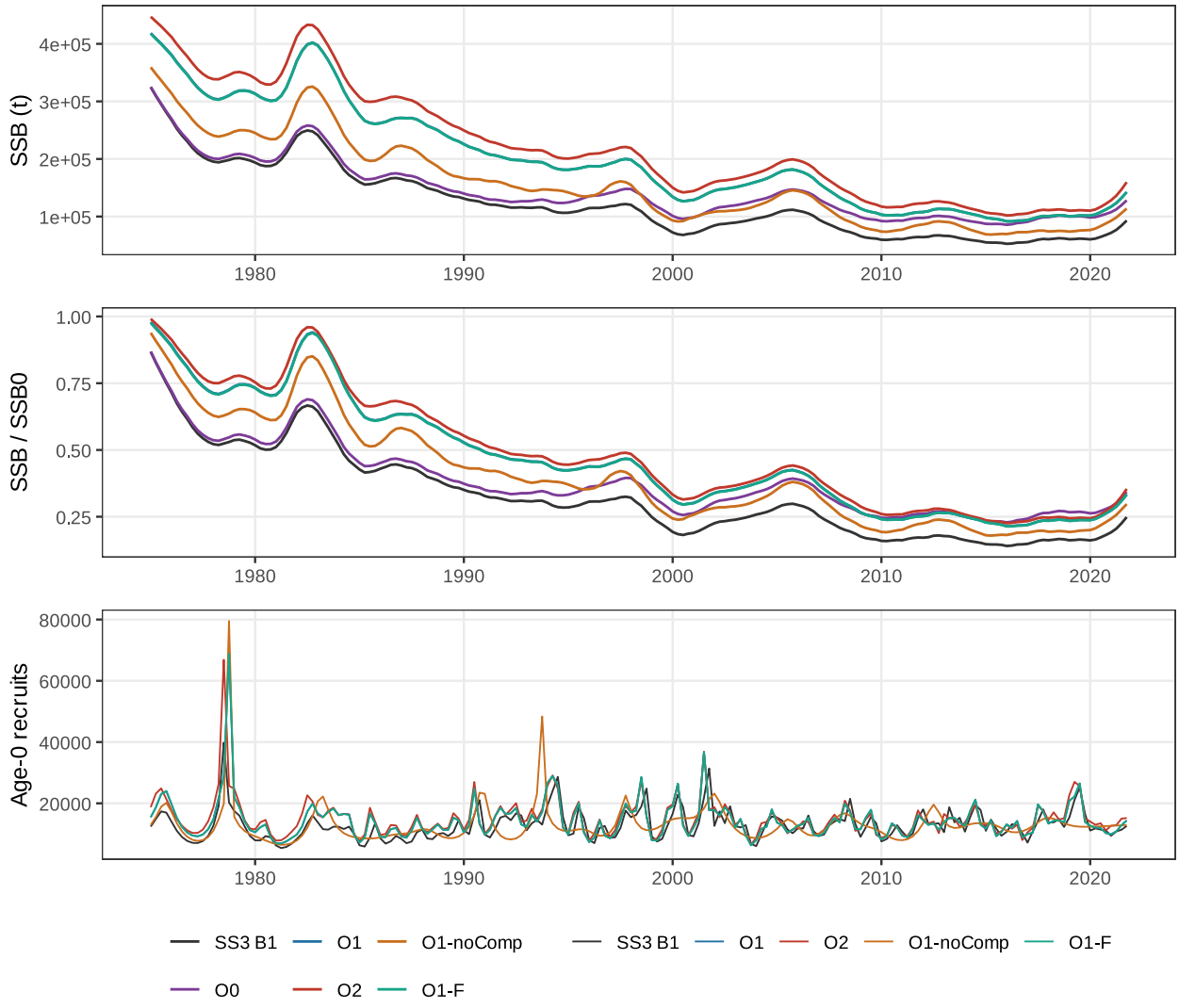


Figure 5: Spawning biomass, depletion and age-0 recruitment: SS3 B1 and the opal fits.

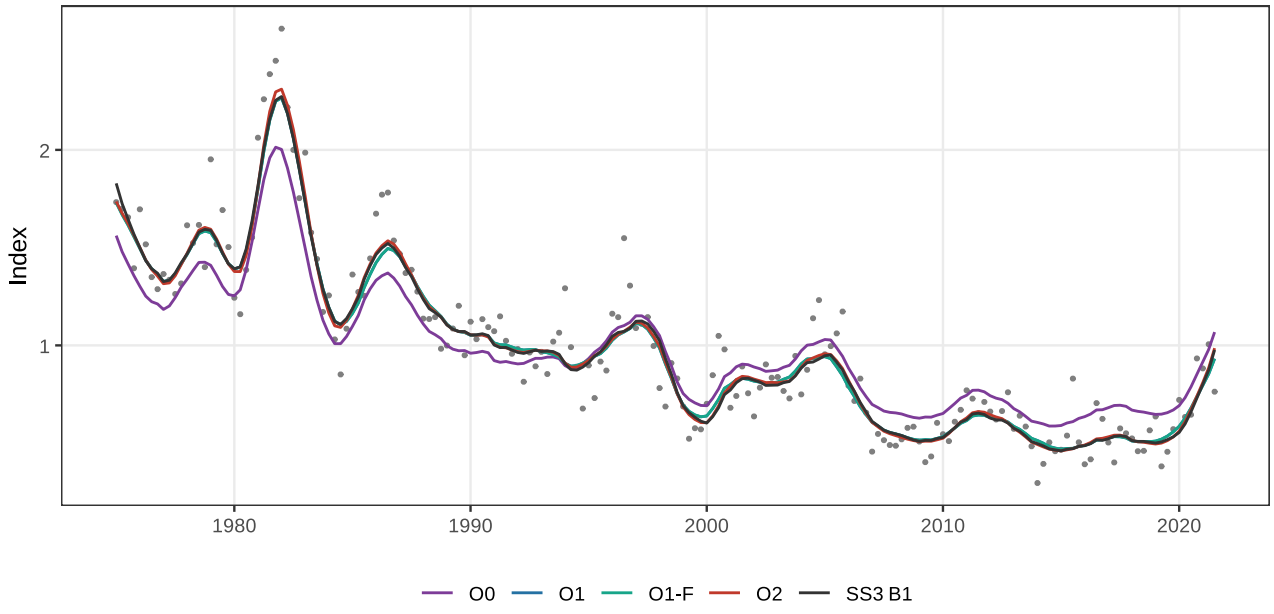


Figure 6: Fits to the longline index (observed: points).

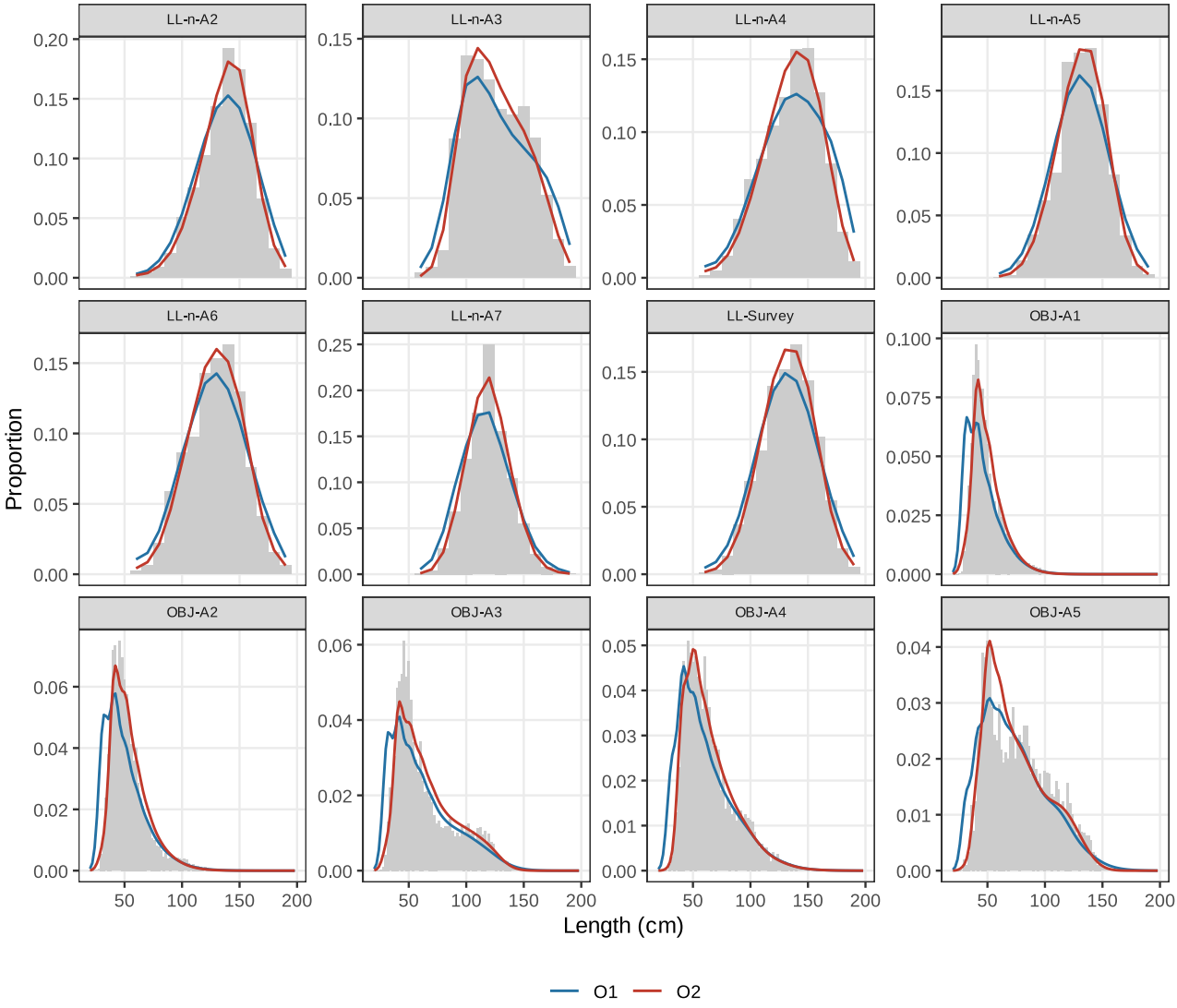


Figure 7: Compositions aggregated over time, by fishery (observed: grey bars): OBJ length compositions (2 cm) and the longline and index size compositions (10 cm).

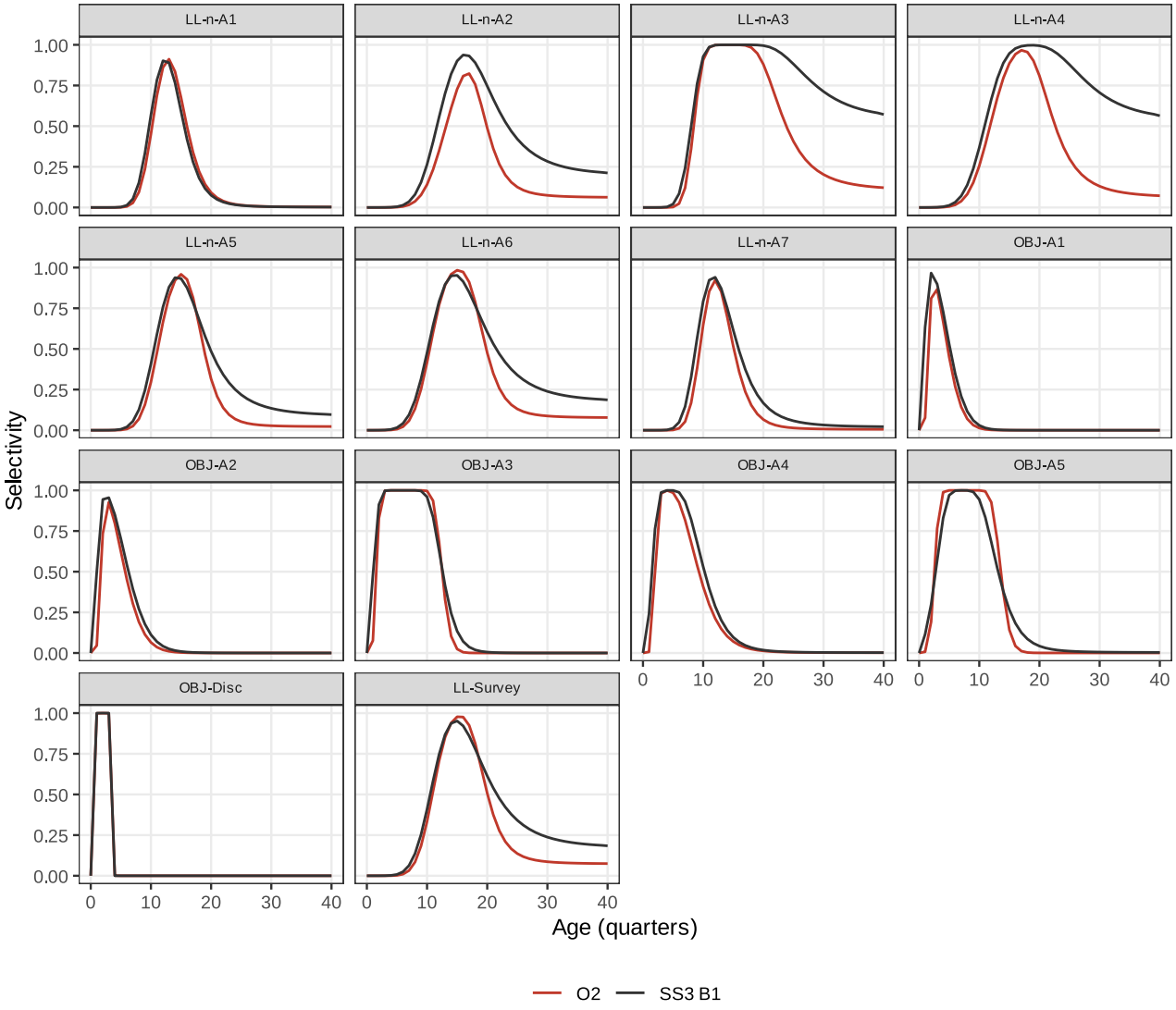


Figure 8: Selectivity-at-age: SS3 B1 and opal O2, by fishery (mirrored fisheries omitted).

6 Estimating the initial F

SS3 starts the population from an equilibrium fished at an initial F for each fleet with an initial equilibrium catch. In this model those are LL-n-A2 and OBJ-A5, with an arbitrary catch of 999 and lambda 0 on the initial-catch likelihood, so their initial Fs are not tied to a catch; the control file fixes them (phase −1) at 3.3e-03 and 1.8e-07 per quarter, an almost unfished start. The regime offset is what depletes the initial state in B0 (Section 3). opal has the same parameter (`log_init_F_f`, one per fishery, applied with each fishery's selectivity in the initial equilibrium), and the conversion carries SS3′s values over as fixed. Neither model has an initial catch in its likelihood, so the initial Fs are informed only by what the initial age structure does to the index and the compositions. Three runs estimate them:

- **B0-F**: B0 (regime offset on) with both initial Fs estimated;
- **B1-F**: B1 (regime offset off) with both initial Fs estimated, the SS3 counterpart of opal (which has no regime offset, so there is no opal run with the offset on);
- **O1-F**: opal O1 with the initial F of the same two fisheries estimated.

Each starts from its parent's estimates, with OBJ-A5′s initial F moved off ~0 to 0.001. O1-F is also started from B1-F's estimates, and both starts reach the same optimum (objectives 6,369.833 and 6,369.833).

Table 8: Estimating the initial F (per quarter; SS3′s SE in brackets). Initial depletion is the SSB at the start of 1975 Q1 relative to unfished; final depletion is in 2021 Q4. SS3′s NLL and opal's objective are not comparable with each other; compare within each model.

model	regime offset	initial F LL-n-A2	initial F OBJ-A5	NLL / objective	max gradient	Hessian PD	SSB0 (t)	initial depletion	final depletion
B0	−0.409	3.3e-03	1.8e-07	271.20	1.8e-03	TRUE	371,192	0.649	0.221
B1	off	3.3e-03	1.8e-07	274.60	2.0e-02	TRUE	374,298	0.868	0.249
B0-F	−0.407	4.0e-03 (7.2e-02)	1.9e-07 (1.9e-04)	271.20	1.2e-04	TRUE	371,169	0.647	0.221
B1-F	off	2.9e-07 (2.9e-04)	4.5e-02 (2.2e-02)	272.40	4.3e-03	TRUE	369,021	0.640	0.226
O1	none	3.3e-03	1.8e-07	6,369.85	1.9e-11	TRUE	427,626	0.979	0.333
O1-F	none	2.0e-09	3.4e-03	6,369.83	3.5e-06	TRUE	427,943	0.976	0.333

With the regime offset on (B0-F) the initial Fs add nothing: the likelihood changes by 0.000 units, the offset stays at −0.41, OBJ-A5′s initial F returns to its lower bound and LL-n-A2′s moves to 4.0e-03 with an SE of 7.2e-02, many times the estimate. With the offset present the data cannot separate a lower initial recruitment from a fished initial equilibrium, so the initial F is not estimable alongside it.

Without the offset (B1-F) the initial F does much of the offset's job. The likelihood improves by 2.20 of the 3.40 units lost by removing the offset, through OBJ-A5′s initial F (4.5e-02 per quarter, SE 2.2e-02), while LL-n-A2′s goes to its lower bound. The initial depletion falls from 0.868 (B1) to 0.640, against 0.649 in B0, and the 2021 Q4 depletion is 0.226 against 0.221 (B0) and 0.249 (B1). The initial F also replaces most of the negative early recruitment deviations B1 uses to lower its initial state (their mean is −0.08 in B1-F against −0.24 in B1). B1-F's initial numbers-at-age match B0′s within 4% from age 8 quarters, where B1′s are up to 52% above them. The two mechanisms still differ at the youngest ages (Figure 10): the offset lowers every initial age class by the same factor, while the effect of an initial F builds up with the number of quarters a cohort has been fished, so B1-F's youngest ages stay above B0′s. What the initial F does not recover is the remaining 1.20 likelihood units.

In opal (O1-F) the fit converges and passes opal's fit check. The initial Fs are 2.0e-09 (LL-n-A2) and 3.4e-03 (OBJ-A5) per quarter, an order of magnitude below B1-F's for OBJ-A5, and the objective improves by only 0.014 units on O1. The initial and 2021 Q4 depletions are 0.976 and 0.333 (O1: 0.979 and 0.333), so in opal the initial F changes almost nothing: O1 already starts from a nearly unfished state, and the compositions that push O1 to a larger, less depleted stock (Section 5) give no reason to deplete the initial state. The initial F is therefore not a route to reproduce B0′s initial state in opal while the catch-timing difference remains. opal applies the initial F as a harvest fraction at the start of the quarter, 1 - exp(-F), before natural mortality, where SS3′s initial equilibrium uses its catch equation; at these levels the difference is small next to the catch-timing difference of Section 4.4, which O1-F inherits.

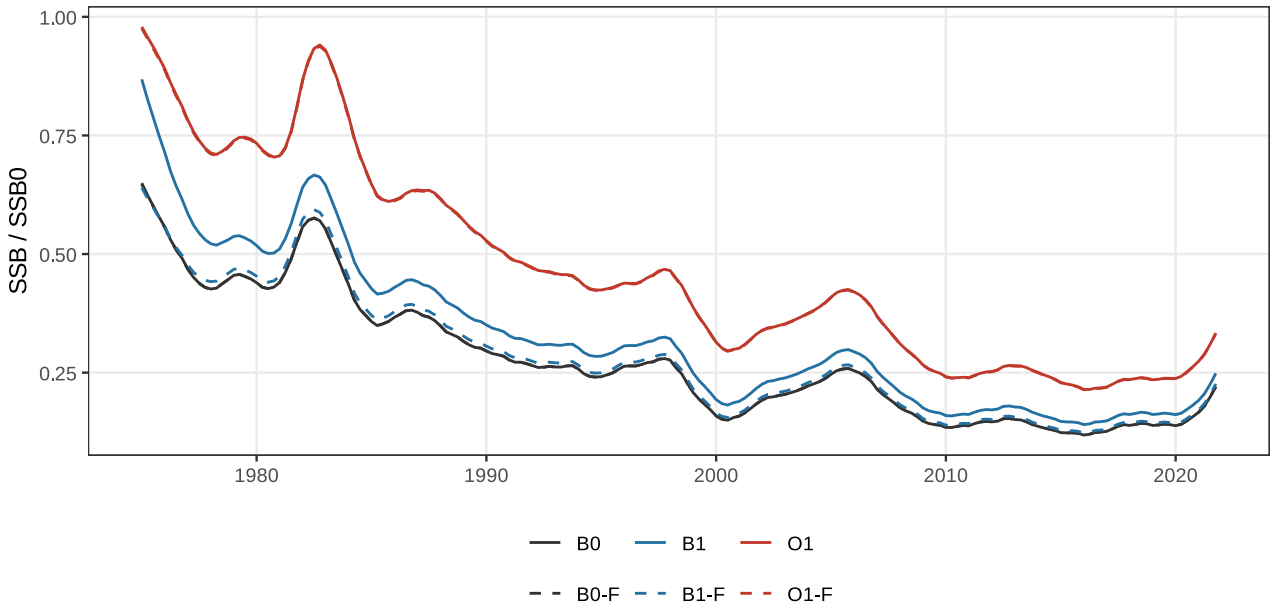


Figure 9: Spawning biomass relative to unfished: the SS3 runs with the initial F fixed (B0, B1) and estimated (B0-F, B1-F), and opal O1 and O1-F.

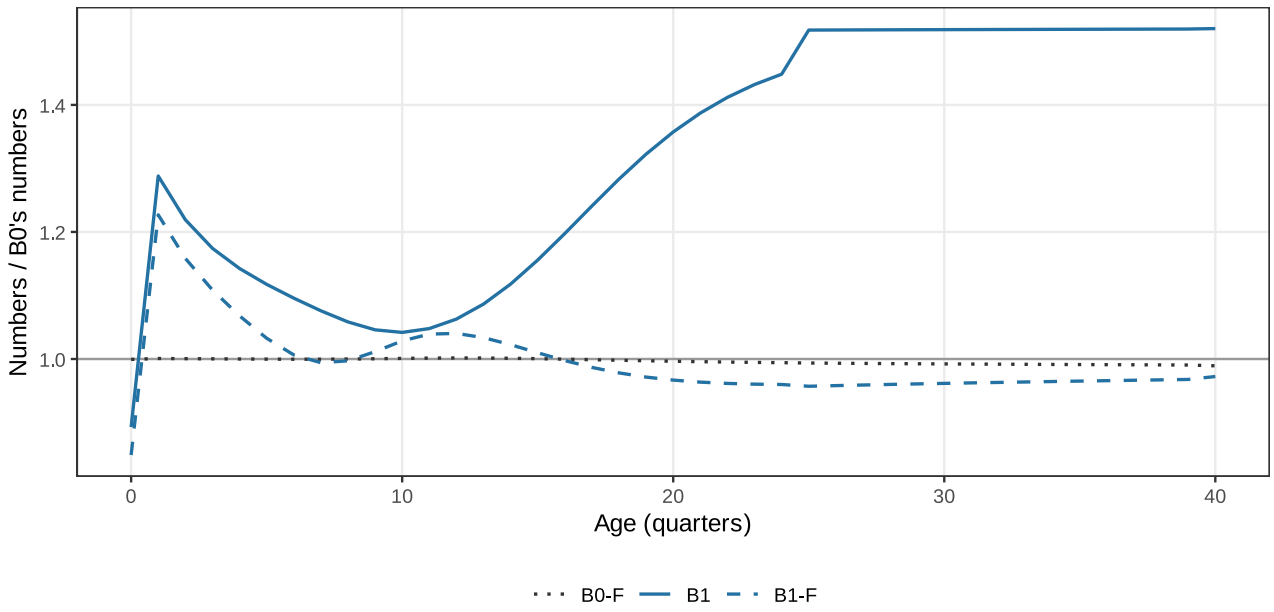


Figure 10: SS3's numbers-at-age at the start of 1975 Q1 relative to the supplied model's (B0). Without the regime offset (B1) the older ages start about 50% higher; with the initial F estimated (B1-F) they are close to B0's again. B0-F is indistinguishable from B0.

7 Limitations and differences from SS3

7.1 Not available in opal

- **Within-step catch timing.** opal removes each time step's catch at its start. How much that matters depends on the mortality within a step (natural and fishing) and on how young the selected fish are, not on the step length alone: it is small when both are low. For this stock, with M of 0.48 per quarter at age 0 and large catches of young fish, it is the largest difference (Table 6). A mid-step removal (Pope) would recover most of it, and SS3's catch equation nearly all of it.
- **Fishery-specific catch weight-at-age.** opal turns catch in weight into numbers with one weight-at-age vector for every fishery, at one time in the step. SS3 uses each fleet's catch weight-at-age, which accounts

for growth through the step and for size selection within an age. A fishery-by-age catch weight (like the external selectivity matrix) would carry SS3′s exactly.

- **Initial-equilibrium recruitment offset.** SS3′s regime offset scales the initial equilibrium recruitment, unpenalised; opal has no equivalent, so B1 removes it, which moves the whole trajectory (Section 3). opal's initial deviations (`init_rdev_a`, one per initial age class) could carry B0′s initial numbers-at-age exactly at SS3′s values, but opal penalises them with sigma_R (about 10 likelihood units for an offset of −0.41 on 41 ages), so a fit would pull the initial state back towards unfished. An unpenalised offset on the initial equilibrium (one parameter in opal's initial numbers) would let opal be converted from B0 directly. Estimating the initial F (B1-F, O1-F; Section 6) recovers part of what the offset does, with a different initial age structure.
- **Age selectivity combined with size selectivity.** SS3 multiplies each fleet's length-based curve by an age curve (pattern 10: 0 at age 0; pattern 11 for the discards); opal has one or the other per fishery.
- **Conditional age-at-length data** (a few records, with lambda 0 in SS3).
- **A complete decomposition of the objective.** With the compositions in the likelihood, opal's objective is larger than the sum of the weighted terms it reports (by 2,302 units for O1; Table 7), and the difference disappears when the compositions' lambda is 0, so part of what the composition likelihood contributes is not in the reported `lp_lf`. The NPO blue shark fits show the same. Until it is resolved, the reported terms cannot be added up to the objective, and composition likelihood comparisons between fits rest on the reported `lp_lf` only.

7.2 Other structural differences and conversion choices

- **Plus-group length.** SS3 averages the plus group's length over the ages it accumulates; opal uses the growth curve at the oldest age (0.09 cm).
- **Observation timing.** SS3 predicts the fishery compositions from the catch through the quarter, with mid-quarter lengths, and the index in month 1; opal predicts both from numbers and lengths at the start of the quarter.
- **Generalised size compositions** become opal's second composition bin scheme, with SS3′s variance adjustments (mult_by_generalized_sizecomp).
- **Catchability.** SS3 solves it analytically (floating q); opal estimates it.
- **Late recruitment deviations.** SS3 estimates deviations for the two quarters after its last main deviation with the same sigma_R penalty; opal estimates them as ordinary deviations.

Reproducing this document

Step	Script	Writes
1	`scripts/01-build-ss3-bridge.r`	the SS3 bridge models and `vignettes/data/epo-bet-ss3-bridge-cache.rds`
2	`scripts/02-convert-fit-epo-bet-opal.r`	`vignettes/data/epo-bet-opal-cache.rds` and the fitted models
3	`quarto render vignettes/epo-bet-ss3-to-opal.qmd`	this document

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Caches: bridge 2026-11-03 02:09:57 UTC (SS3 3.30.22.1); opal 2026-11-03 02:09:12 UTC (opal
0.1.2, r4ss 1.56.0, R 4.5.3).
```

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[2] IATTC, "Exploratory stock assessment of bigeye tuna in the eastern Pacific Ocean," La Jolla, California, USA, Document prepared for the 14th Meeting of the Scientific Advisory Committee (SAC-14), 2023.

[3] M. N. Maunder *et al.*, "The growth cessation model: a growth model for species showing a near cessation in growth with application to bigeye tuna (Thunnus obesus)," *Marine Biology*, vol. 165, p. 76, 2018, doi: 10.1007/s00227-018-3336-9.